

Cause and Effect

Go wide across the categories before you go deep, so that a whole class of cause is not simply forgotten.

BEST FOR A PROBLEM, A RISK · KAORU ISHIKAWA, 1960S · TAKES FORTY MINUTES

HOW TO RUN IT

1. Write the effect as something observable: what happened, when it was first seen, and how you know. Not how it felt.
2. Draw six ribs and name them: people, process, technology, materials and inputs, environment, measurement. Adapt the names if yours are obviously different, but keep six.
3. Work every rib in turn, and list the causes that could sit under it, one per line, each stated as something that happens or fails to happen. Force at least one weak cause into the ribs you would have skipped.
4. Beside each cause, write what makes you think so: a log, a message, a thing you saw, a person who saw it. Then mark it verified, suspected or ruled out.
5. For every cause marked suspected, write the one thing you would look at to check it.
6. Name the single cause you would bet on, say why in terms of the evidence rather than the conviction, and write the one test that would show you were wrong.

Where it falls down. The default categories don't suit every problem, and people either force causes into headings that don't fit or skip the headings that feel irrelevant.

WITH A PEN

The effect

What happened, how you know, when first seen

People

Cause / what makes you think so / verified, suspected, ruled out

Process

Technology

Materials and inputs

Environment

Measurement

The one people forget

The most likely cause

Why that one, in evidence. The test that would settle it